Drag Reduction System (DRS) in Formula 1

Introduction

Overtaking in Formula 1 racing presents a serious challenge for both drivers and engineers due to the intricate balance between speed and aerodynamics. In the current era of F1, cars are designed for maximum performance in corners, which creates significant aerodynamic challenges when trailing other cars, impacting the ability to overtake effectively. To address this issue, the Federation Internationale de l'Automobile (FIA), the governing body of Formula 1, introduced the Drag Reduction System (DRS).

DRS is a regulated aerodynamic mechanism that temporarily reduces rear wing drag to increase a car's straight-line speed under specific race conditions. The system works by coordinating track-based detection sensors, onboard electronics, and an adjustable rear wing flap. This adjustable flap changes the way air flows over the wing, allowing for a temporary speed advantage which can increase overtaking potential. Understanding how DRS functions requires examining the aerodynamic forces that affect F1 cars, the mechanical components of the system, and the sequence of events that allows it to operate during a race.

Aerodynamics

To fully grasp how DRS functions, it is necessary to define key aerodynamic terms that can influence Formula 1 car performance.

Drag is the aerodynamic resistance that opposes a vehicle's forward motion as it travels through the air (*Figure 1*). This force is like the resistance felt when sticking a hand outside a moving car window. Drag increases with speed, which limits top velocity. DRS addresses this issue by temporarily reducing drag.

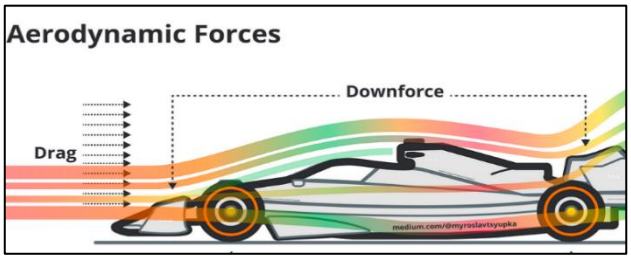

Figure 1: Simplified airflow over car

Downforce is the vertical aerodynamic force that presses a car down to the track (*Figure 1*). This force is generated by the wings and underbody of the car. It is like pressing down on a toy car while pushing it around a corner to prevent sliding. Downforce is important through high-speed corners as it keeps the car's tires gripped to the track. While beneficial in turns, downforce increases drag, creating a balancing issue that DRS was introduced to manage.

Dirty Air refers to the turbulent airflow created behind a moving car. Similar to the wake left behind a moving boat, the disturbed air reduces aerodynamic efficiency for trailing vehicles (*Figure 2*). DRS provides the trailing car with additional speed on straight sections, which compensates for the aerodynamic disadvantage caused by dirty air; this creates overtaking opportunities, leading to more exciting racing.

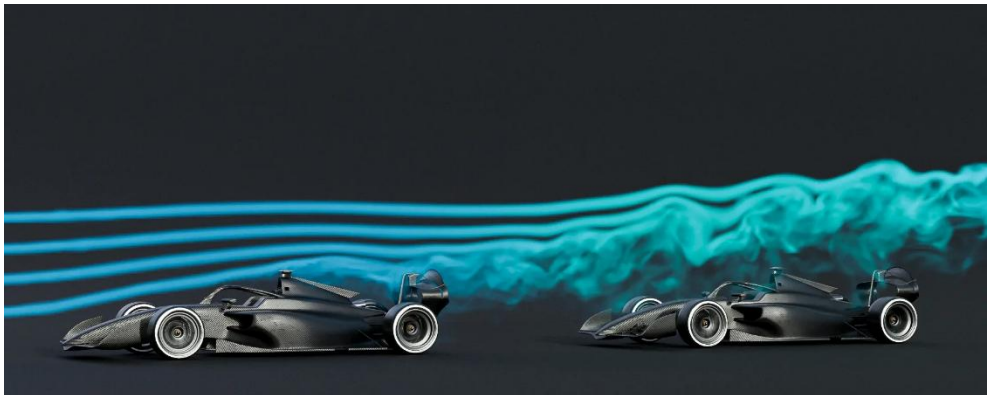

Figure 2: Visualization of dirty air for trailing F1 cars

Structural Components

DRS consists of four primary components: the rear wing assembly, an electronic actuator, the driver control interface, and an FIA-installed timing system.

Rear Wing Assembly

The rear wing assembly can be broken down into two parts: a fixed main plane and an adjustable upper flap. In its standard configuration, the flap is closed, forming a narrow gap that maximizes downforce. As seen in *Figure 3*, opening the flap widens the slot gap, reducing aerodynamic resistance.

Figure 3: Rear wing open vs closed comparison

Electronic Actuator

The electronic actuator is used to control the movement of the upper flap. When triggered, it rotates the flap into the open position and returns it to the closed position automatically when required.

Driver Control Interface

The driver control interface is the system a driver uses to manually activate DRS. A steering wheel button sends a signal to the actuator to open the flap. However, the flap only opens if eligibility conditions are met.

Track Detection Systems

Track detection systems are installed throughout the race circuit. These measure the time interval between cars at detection points around the track. DRS can only be opened if a trailing car is within one second of the car ahead in designated activation zones. It will automatically close if the trailing car falls out of the one second gap or if the end of the DRS zone is reached.

Operation

The operation of DRS uses the following sequence.

1. **Time Gap Detection:** At the detection zone, track sensors measure the interval between two cars. If the gap is one second or less, it becomes eligible to use DRS. This rule ensures DRS assists cars already close enough to attempt an overtake. These points are illustrated in *Figure 4* with the track sections labelled "DRS detection".

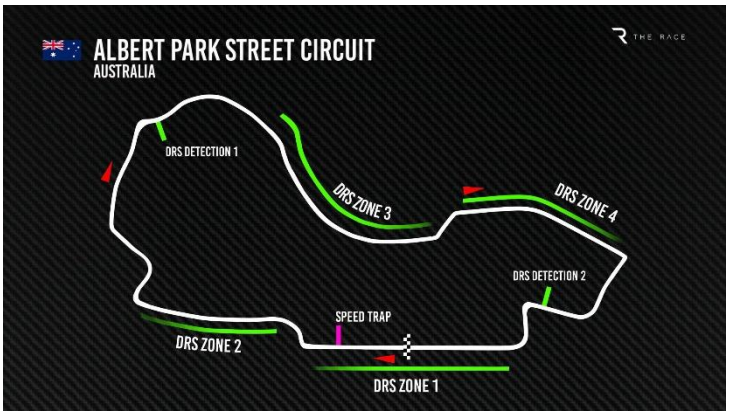

Figure 4: DRS detection zones and straights on an F1 circuit

2. **Activation Zone Entry:** Upon entering a designated DRS straight, the car's onboard system allows DRS activation. The zones are typically placed on long straights where overtaking is most likely. These DRS straights can be seen in *Figure 4*, labelled as "DRS Zone".
3. **Driver Input:** The driver presses the steering wheel button, sending a signal to the actuator.
4. **Flap Adjustment:** The actuator pivots the upper flap upward, to open the flap and reduce drag. Opening the flap changes the rear wing's aerodynamic profile, reducing drag and increasing straight-line speed.
5. **Speed Increase and Closure:** Reduced drag allows greater straight-line speeds. When the driver brakes or exits the zone, the system automatically closes the flap, restoring full downforce.

Impact and Competitive Purpose

In the current era of Formula 1 regulations, DRS plays a key role in racing strategy. Reducing drag in straight sections allows for a speed advantage that increases overtaking chances.

Because overtaking is a key component of competitive racing, increasing passing opportunities leads to more engaging and unpredictable races for both teams and spectators.

Since DRS provides a significant advantage, it is tightly regulated. The FIA disables the system in wet conditions and during safety car periods to maintain safety and fairness. Additionally, as mentioned previously, DRS can only be used in predetermined activation zones around the track. These restrictions ensure that the system enhances competition without allowing an unrestricted advantage.

DRS is used as both a mechanical system and as a form of regulation designed to allow engineers to innovate while also maintaining fairness in the sport.

Conclusion

The Drag Reduction System is a regulated aerodynamic system designed to temporarily reduce rear wing drag under certain race conditions. It operates through a complex combination of manual and automatic systems that alter airflow to increase speed on straights. Although DRS has generated debate regarding its influence on competitive balance, it remains an integral part of modern Formula 1 racing. The new FIA regulations for 2026 introduces Active Aerodynamics. This new system builds on DRS by adding an adjustable front wing flap in addition to the rear wing. This demonstrates how DRS has become a foundational technology within Formula 1, shaping the future of aerodynamic design in the sport.

Image Citations:

Figure 1: https://medium.com/@darienjy5056/will-cfd-technology-shape-the-next-era-of-f1-aerodynamics-6b39b9a3820b

Figure 2: https://www.raceteq.com/articles/2025/07/slipstream-vs-dirty-air-explained

Figure 3: https://medium.com/@donovan.brown/understanding-drs-in-formula-1-e3b78a5826bf

Figure 4: https://www.the-race.com/formula-1/f1-cuts-melbourne-drs-zones-from-four-to-three/